



COMPONENTS

High-bandwidth op-amp

The TSV7722 precision high-bandwidth operational amplifier, with gain-bandwidth of 22MHz and slew rate of 11V/ μ s, is well-suited for high-speed signal conditioning and accurate current



measurement in power-conversion circuits and optical sensors. It

enhances energy efficiency in applications such as automotive smart mobility, solar panels, telecom infrastructure, and computer servers. Besides maximum input-offset voltage of 200 μ V and extremely low input-voltage noise density of 7nV/ $\sqrt{\text{Hz}}$, an input bias current of 2pA enables accurate photodiode-current measurement in optical-sensing applications such as smoke detectors. Unity-gain stable and fully specified on a load of 47pF, the TSV7722 can be used as an input buffer for analogue-to-digital converters (ADCs).

STMicroelectronics

www.st.com

Satellite image processing SRAM

The 144Mb quad data rate II+ (QDR-II+) SRAM is a high-speed external cache memory ideal for radar, on-board data processing, and networking



applications in space. Unlike DRAM with higher latency and memory bank restrictions, the new SRAM

reduces the overall system complexity in radar and imaging applications by delivering higher performance that enables on-system satellite image processing with better resolution and faster processing speeds. It operates up to a maximum frequency of 250MHz, delivering up to 36Gbps throughput. It is radiation-hardened for up to 200 krad (Si) and is latch-up immune, improving

system reliability in harsh environments.

Infinion Technologies

www.infineon.com

High-power shunt resistors

Designed for high-power applications, the newly developed GMR320 series of shunt resistors featuring a rated power of 10W is suitable for automotive, industrial equip-



ment, and home appliances. It offers a resistance value ranging from 5m Ω to

100m Ω . Unique structure and optimised materials allow the GMR320 series to reduce surface temperature rise by 23%, ensuring high ruggedness against overcurrent loads. In addition, a high performance metal alloy for the resistive material provides low temperature coefficient of resistance that makes reliable and highly accurate current detection possible even in the low resistance values.

ROHM Semiconductor

www.rohm.com

Touchscreen controller

Simplifying system development for ultrawide screens often seen in electric vehicles (EVs), advanced driver-assistance systems (ADAS), and premium vehicles is the new max-



Touch MXT2912TD-UW touchscreen controller.

This automotive-qualified, single-chip solution addresses display sizes up to 114cm (45 inches) with a very wide aspect ratio, supporting liquid-crystal display and OLED display technologies. It reduces the need for multiple touch controllers within a vehicle's human-machine interface display. It also enables detection and tracking of multi-finger touch through thick gloves and a wide variety of overlay materials and thicknesses, even in the presence of moisture.

Microchip Technology

www.microchip.com

ESD protection diode

The new bidirectional symmetrical 2-line ESD protection diode VBUS05M2-HT5 offers ultra-low capacitance and leakage current for the protection of



high-speed data lines against transient voltage signals. Offering low load capacitance

of 0.37pF typical, the device is optimised for interface protection including USB 2.0, USB 3.0, and HDMI in commercial applications. The protection diode features a low maximum leakage current of <0.1 μ A at a working voltage of $\pm$ 5.5V, a breakdown voltage of 8.5V typical at 1mA, and a maximum clamping voltage of 18V at 3.4A. RoHS-compliant, halogen-free, and Vishay Green, it supports automated optical inspection for automotive systems.

Vishay Intertechnology

www.vishay.com

Integrated multicore processors

To satisfy demands from military and avionics sectors, the QorIQ T4240 processors can address extremely compute-intensive application scenarios. Comprising up to 12 dual-threaded PowerArchitecture cores (24 virtual cores) and attaining fre-



quencies of 1.8GHz, these deliver elevated levels of performance while taking

up only minimal board space. The processors feature an extensive range of I/O options including 10Gbit Ethernet, 1Gbit Ethernet, PCIe, USB, SATA, and RapidIO. Thanks to the data path acceleration logic embedded into each T4240, packet classification and processing efficiencies can be boosted significantly. Operational temperatures spanning from -55°C to 125°C are supported.

Teledyne e2v

www.teledyne-e2v.com

High flexibility microcontroller

NuMicro M0A21/M0A23 series is a 32-bit Arm Cortex-M0 high-performance micro-